
Detecting Large Language Models writing

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Abstract

We develop a set of models to detect text generated by Large Language Models (LLMs) using a dataset from Kaggle that comprises labeled text originating from multiple LLM and human sources. After describing a set of stylized facts about what separates LLM from human writing, we start by fitting a set of logistic classifiers based on simple heuristics. We then use a pre-trained version of DistilBERT, a lighter version of the bigger BERT transformer, to compare the performance with our simpler benchmark models. We find that the latter are much closer to DistilBERT than expected.

1 Introduction

In a study published in 2025, OpenAI reported that "Writing" figured as one of the three most common conversation topics with ChatGPT, alongside "Practical Guidance" and "Seeking Information". While some use cases may be benign, such as writing emails for work, or double-checking the draft of a report, there have been many documented cases of AI being used for harmful purposes. For instance, it has made diffusing fake news easier since you can now generate plausible-looking content at scale. It has also made it more difficult to find reliable and quality information on forums and blogs because these platforms are now flooded as well with AI-generated messages and articles. Of course, it also has served students to accelerate the completion of homeworks with dire consequences for their learning.

This has pushed those in the machine learning community to develop tools capable of discriminating between AI-generated and human-written texts. While this goal seemed elusive for some time, there are now pre-existing solutions with some of them being very successful such as Pangram which boasts a 99.98% accuracy score on ChatGPT, Claude, Gemini and more. Other freely available platforms also exist but with varying degrees of reliability. Many of these solutions use creative techniques to discriminate the text. Because it is a timely and highly relevant topic, we thought that it would be interesting to build our own LLM-writing detector using simpler models and see how far they could take us.

We start this report with a literature review showing the state-of-the-art techniques used in traditional AI detectors. We then move on to presenting the datasets that we leverage for modeling. After that, we will present our methodology, and end up with a presentation and a discussion of the results.

2 What the literature says

Common methods

Fraser et al. (2025) classify AI-generated text detection methods into three broad families: watermarking, statistical and stylometric analysis, and supervised classification with pretrained language models. Watermarking methods put a hidden signal during generation, but require cooperation from the model provider and may be vulnerable to paraphrasing or editing. Statistical methods exploit consistent patterns in the generation process, using measures such as token probability, rank, entropy, perplexity, or probability curvature, as in GLTR and DetectGPT, which are two off-the-shelf AI detection tools. Stylometric methods instead rely on observable linguistic properties such as lexical diversity, repetition, punctuation, readability, syntactic structure, and coherence. Finally, pretrained language-model classifiers such as BERT- or RoBERTa-based detectors learn detection patterns directly from labeled human and AI text. The authors emphasize that the effectiveness of these approaches depends strongly on the detection setting, including whether the generator is known, whether probability access is available, the domain and language of the text, document length, and the degree of human editing.

On stylistic differences between LLMs and humans

Reinhart et al. (2025) find that LLMs do not fully write like humans, even when they are prompted to imitate human style. Their writing has grammatical and structural patterns that can indeed be detected. Looking at GPT-4o, Llama and other mainstreams LLM models, they find that these models tend to overuse nominalizations, present participial clauses¹, "that" clauses as subjects and phrasal coordination² relative to human writers. They also document differences across LLMs; for instance, that GPT4o uses the passive voice at roughly half the human rate while Llama uses it roughly on par with humans.

Another paper from Guo et al. (2023) compare the differences between human answers and GPT3.5 answers to a large set of questions. Their findings are numerous. ChatGPT answers tend to be a lot more structured by first defining the concept, then giving a step-by-step answer and finally by ending with a summary of the answer. They also document that the LLM answers are usually longer, but yet its vocabulary is less diverse than that of humans. They then explain that humans tend to be more expressive with a higher density of exclamation marks, question marks and ellipses. On the other hand, ChatGPT relies a lot more on logical connectors and adverbs like "Firstly", "Secondly" or "Finally". And lastly, they find that ChatGPT-generated texts have lower perplexity, suggesting that they are more predictable and closer to common language-model patterns than human writing.

3 Data

3.1 General presentation

For this task, we will rely on a dataset proposed on Kaggle, a data-science competition online platform, which was made available specifically for the task of training AI-text detection algorithms. This dataset contains "*a mix of student-written essays and essays generated by a variety of LLMs*". We think that this will be beneficial to the performance of our models to the extent that it will not overfit on the output of a specific version of an LLM model.

To measure the true performance of our models in practice, we will also leverage another supplementary dataset. It comes from the GitHub repository of Tom Li, a contributor on a blogging platform called Medium, who shared the data that he used to test his own AI detection algorithms using BERT-type models. Just like our own training data, it comes from a variety of LLM sources including

¹A clause built around a verb ending in -ing

²When two or more phrases are linked using connectors like and, or, or but.

ChatGPT, Claude, Gemini, Deepseek and more. We therefore think that it is a suitable external data to test our models. An interesting feature of this data is that we have access to the source, so we will be able to compute the performance statistics for each LLM source. We expect this to be informative about the heterogeneity in LLM writing style. For the human texts in the data, they come from the British newspaper The Guardian and from the abstracts of scientific articles available on ArXiv.

Table 1 below shows a recap of the data that we use, with a breakdown of LLM sources when relevant.

Table 1: Meta-information on the various datasets used

	# Human Answers	# LLM Answers	Source
Kaggle data (total)	15 823	10 465	Kaggle challenge ¹
Medium data (total)	1 213	3 787	Medium website ²
GPT-4		1 185	
Claude		1 320	
Gemini		664	
Deepseek		349	
Llama		180	
Mistral		189	

3.2 Descriptive statistics

In this subsection, we add some descriptive statistics to show how AI and human texts can be different on stylometric-only grounds. For this, we will use the training set of our Kaggle data. We focus on a few arbitrary features here which will be the average sentence length, the lexical diversity, the repetition rate of words, and the standard deviation of word lengths. We define lexical diversity as the number of unique words over the total number of words in a text. The repetition rate is the share of words that are repeated more than once in a given text. Therefore, both metrics lie between 0 and 1. Table 2 shows the calculated metrics averaged at the AI and human level.

Table 2: Descriptive stylometric statistics by text origin

Text origin	Average sentence length	Lexical diversity	Repetition rate	Word length std.
AI	22.83	0.559	0.605	2.73
Human	22.32	0.443	0.724	2.26

We can make a few remarks here. First, it seems that AI writes slightly longer sentences but it is barely noticeable. In this sample of texts, humans seem to enjoy using a smaller set of words with a lexical diversity rate that is 20% lower, and also enjoy repeating this set of words more often with a 19.5% higher repetition rate. That comforts us with the TF-IDF approach to logistic regression which we expect to identify those specific markers of human speech. For some reason, AI also seems to use words that are more varied in length size, with those figures implying a variance that is 46% higher.

We also performed part-of-speech tagging which consists in assigning each word in a text to a specific grammatical category such as noun, verb, adjective and more. Therefore, it captures better the grammatical structure of the texts than the simple metrics that we showed above. We use the

¹ You can find Kaggle data here: <https://www.kaggle.com/datasets/sunilthite/llm-detect-ai-generated-text-dataset>

² You can find the Medium data on this GitHub: <https://github.com/tommyliphysics/athena-source>. We used the test.csv file in the data folder.

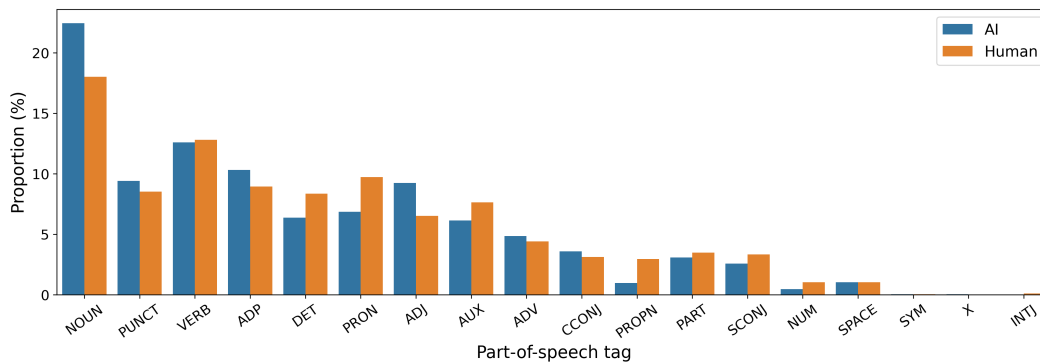


Figure 1: Part-of-speech comparison

Python library SpaCy to compute for each text the proportion of words belonging to each grammatical category, and then we average those proportions for AI-generated and human-written texts.

The results of the tagging in Figure 1 show clear differences in grammatical composition. AI texts contain a much larger share of nouns which Nagy et al. (2012) interpret as a sign of more argumentative texts, and we also have a larger share of adjectives, adpositions (which include prepositions) and adverbs. Taking all this together, we may conclude that this is a sign of a more formal and descriptive writing style. In contrast, human texts have a relatively larger share of determiners ("the", "a"), pronouns, auxiliary verbs ("is", "have"), proper nouns and subordinating conjunctions ("because", "although"). So human-written texts seem more personal and more context-based to the extent that they make more references to specific people or entities, and refer back to them with pronouns.

So overall, this confirms that both the distribution of words and the grammatical structure of texts can help us discriminate between AI and human writings.

4 Methodology

We will adopt a multi-layered approach to modeling. As we announced in the introduction, we will first develop a basic benchmark against which we will later compare more complex models. The idea is to see how far a model based on simple heuristics can take us. Our first model will be a logistic regression classifier based only on the TF-IDF features of the training sample. We pick it as our starting model because it is straightforward to implement, and requires little oversight on defining the features that the model will be using to discriminate the texts. As we described above, LLMs and humans tend to use different sets of words. It has been documented in particular that LLMs have "preferred" words such as *delve* or *underscore* when it comes to ChatGPT as anyone who used it repeatedly can tell. We expect this simple benchmark to be able to capture those differences in the distribution of words employed and have decent performance as a start-off.

We then train two other logistic classifiers, with arguably more advanced features. First, we will measure a set of stylistic features based on what is recommended in the literature and our own experience of using LLMs daily, to feed a logistic classifier. The features that we retained are described in Table 3. We will then fit another logistic regression that combines the TF-IDF and stylistic features to see how these models compare to the simpler benchmark developed above. We expect the differences in these models' performances to be informative about the heterogeneity in LLMs' writing style.

We also train a transformer-based classifier using DistilBERT. DistilBERT is a lighter version of BERT designed to reduce computational cost while preserving most of BERT’s language understanding performance. In contrast with TF-IDF and stylometric models, DistilBERT does not rely on manually constructed textual features. Instead, it will directly process the raw text and build contextual representations, so that the meaning of each token is modeled conditional on its surrounding words.

The pre-trained distilbert-base-uncased model is fine-tuned for binary classification. Texts are first tokenized with the DistilBERT tokenizer, then truncated to a fixed maximum length in order to save on the training time³. The tokenized inputs are fed into the transformer, and a classification head maps the final representation into predicted probabilities for the two classes.

Data processing

On matters of data processing, we had a minimal approach to data cleaning due to the nature of the task that we perform. In fact, it is usually common in NLP to remove so-called stopwords that carry little semantic meaning such as pronouns or articles. However, important channels through which we expect to detect LLM texts are different distribution of words or stylometric patterns. Therefore, we risk losing essential information if we process our raw data too heavily.

So we took only two steps. First, we removed all duplicated texts to avoid that they hold an unjustified higher weight in the training sample, or that they contaminate the testing or validation sets. There were around 1500 duplicated texts that we removed from the Kaggle data. Then, we removed a handful of texts that were *obviously* AI such as those who were "templates" such that you would have placeholders like "[TYPE YOUR NAME]" or "[TYPE YOUR ADDRESS]". We did not want our models to pick up those artifacts of the training data.

4.1 Implementation of the logistic regressions

We used the typical modeling pipeline with logistic regressions. We split the Kaggle data in three sets: one for training, one for validation and one for testing out-of-sample (which we will call the hold-out sample later on). We use the validation set to select the optimal threshold based on the F1 score to balance under- and over-detection. We will expand on this score in the discussion section. After selecting the optimal threshold (0.45 here), we use the test set to evaluate the model. Results are displayed later in the report.

TF-IDF model

TF-IDF (for Term Frequency-Inverse Document Frequency) gives a higher weight to words that appear often in a text but which are uncommon in the rest of the corpus. So we expect this to be useful to detect words that AI tends to overuse. We also expect this to capture some artifacts that are corpus-specific, that is, specific to our dataset. In its implementation, we kept unigrams and bigrams, removed expressions that appeared in only one document, or expressions that appeared in more than 95% of documents in order not to have an exploding number of explanatory variables.

Stylometric features model

Stylometric features were included to capture dimensions of writing style that are not directly measured by TF-IDF representations. While TF-IDF mainly identifies typical AI words and expressions, stylometric variables describe how a text is written: its length, sentence structure, lexical diversity, punctuation habits, grammatical composition, and readability. These features are relevant for AI-text

³We asked BERT to keep the 512 first tokens of each text, and only around 25% of the sample was longer than this limit.

detection because LLM-generated texts may differ from human texts also, and perhaps mainly, in their consistency, syntactic patterns, repetitiveness, and use of standardized phrasing.

In Table 3, we grouped the 50-or-so features that we used into broader categories to make their presentation easier. We captured features which are distinctive of AI writing, such as *n_em_dash* (i.e. the number of —) or human writing such as *n_ellipsis* (i.e. the number of "...").

Table 3: Main categories of stylometric features used in the analysis

Feature category	Examples		Description
Text length and structure	n_characters, n_sentences	n_words,	Capture the overall size of the text and its basic structural organization.
Word and sentence length	mean_word_length, std_word_length, mean_sentence_length, std_sentence_length		Measure average length and dispersion of words and sentences, which reflect writing style and complexity.
Lexical richness and repetition	lexical_diversity, repetition_rate		Describe vocabulary variety and the extent to which words are repeated in a text.
Character-level composition	uppercase_share, stopword_share	digit_share,	Measure the relative use of uppercase letters, digits, and common function words.
Punctuation usage	count_,, count_!, n_em_dash, n_ellipsis	rate_?,	Capture the frequency and relative intensity of punctuation marks, which may differ across writing styles.
Part-of-speech distribution	pos_share_noun, pos_share_verb, pos_share_adj, pos_share_pron		Measure the grammatical composition of the text through the relative frequency of nouns, verbs and more.
Readability metrics	flesch_reading_ease, flesch_kincaid_grade, gunning_fog		Assess the readability and linguistic complexity of the text using standard readability indices.

5 Results and discussion

In this section, we present the results of our models. We will focus on two metrics. The most important one in our view will be the AI recall rate, which answers the question: "*among all texts that are actually AI-generated, what's the proportion that the model actually detects as AI ?*". A natural benchmark against which we will compare the recall rate of our models is the share of AI texts in the sample that we use. Indeed, if 60% of the texts from a given sample are AI-generated, then randomly guessing that a text is AI-generated 60% of the time will achieve a 60% recall rate for AI. But to balance the tradeoff between detecting AI texts and avoiding over-classification as AI, we will also be interested in the F1 score which is defined as $F1 = 2 * (precision * recall) / (precision + recall)$. Therefore the F1 score lies between 0 and 1 and penalizes models that have too many false positive

predictions. For reference, we will also display precision, which answers the question: "when the model predicts AI, how often is it actually correct?".

Table 4: Performance comparison across evaluation datasets

Model	Hold-out sample			Medium dataset		
	F1	AI recall	Precision	F1	AI recall	Precision
Log-reg TF-IDF	98.53%	97.85%	99.22%	91.75%	99.45%	84.60%
Log-reg Stylo	92.84%	92.36%	93.34%	91.54%	89.89%	87.42%
Log-reg Both	95.82%	94.70%	96.97%	92.78%	92.66%	89.08%

5.1 Logistic regression

Tables 4 and 5 present the results of our logistic regression classifiers.

5.1.1 Hold-out sample

For the hold-out sample, all three models achieve strong performance showing that they generalize well to unseen observations drawn from the same distribution as the training data. The TF-IDF logistic regression performs best overall, with an F1-score of 98.53%, an AI recall of 97.85%, and a precision of 99.22%. This indicates that the model both detects most AI-generated texts and very rarely misclassifies human-written texts as AI-generated. The combined model also performs well, even if slightly below the TF-IDF model, with an F1-score of 95.82%. The stylometric model is the weakest of the three on the hold-out sample, but its results remain solid, with an F1-score of 92.84%. We may conclude that the hold-out results show that lexical structure captured by TF-IDF features is highly discriminative in this dataset but stylometric features alone are less informative.

5.1.2 Medium dataset

For the Medium dataset, performance decreases for all models, which suggests that this dataset is more challenging and that models generalize less in truly out-of-sample settings. The TF-IDF model still achieves the highest AI recall, at 99.45%, meaning that it detects almost all AI-generated texts. However, its precision drops to 84.60%, implying that it produces more false positives by mispredicting human texts as AI-generated. This is important because, in AI detection, false accusations can be costly depending on the stakes. The stylometric model has a lower AI recall, at 89.89%, but a higher precision than TF-IDF, at 87.42%. This suggests that stylometric features are less aggressive in classifying texts as AI, but may be more conservative. As before, the combined model offers the best balance on the Medium dataset, with the highest F1-score of 92.78% and the highest precision of 89.08%, while maintaining a strong AI recall of 92.66%.

Overall, the comparison highlights a trade-off between recall and precision. The TF-IDF model is very effective at detecting AI-generated text, especially when the objective is to minimize false negatives. However, on the external Medium dataset, the combined model appears more robust because it improves precision but still keeps a high recall.

As we described earlier, an interesting feature of the Medium dataset is that we know which LLM generated the AI texts. It is therefore interesting to look, for each source of LLM, what share of texts was detected as AI by our models. This is what table 5 shows below.

The TF-IDF model is by far the most consistent at detecting AI texts, whatever the source they originate from. For almost all LLMs, it classifies correctly more than 98% of texts as AI, and even

Table 5: Share of texts predicted as AI by source, for the Medium data

Source	n	TF-IDF	Stylometric	Combined
claude-3-5-haiku	1080	99.91%	95.56%	98.61%
claude-3-7-sonnet	140	95.00%	55.00%	65.71%
deepseek-r1	101	98.02%	100.00%	100.00%
deepseek-v3	248	97.98%	96.77%	97.98%
gemini-1.5-flash	170	99.41%	98.24%	98.82%
gemini-2.0-flash	494	99.39%	89.88%	91.70%
gpt-4o	109	100.00%	89.91%	94.50%
gpt-4o-mini	1076	100.00%	92.57%	93.77%
llama3.1	67	100.00%	74.63%	82.09%
llama3.2	113	99.12%	59.29%	65.49%
ministral-3b	64	98.44%	75.00%	81.25%
ministral-8b	61	100.00%	62.30%	70.49%
mistral-large	64	100.00%	71.88%	79.69%

reaches 100% for a few of them. This is indicative that vocabulary patterns are already highly informative on this dataset. By contrast, the stylometric model is much less stable across sources. It extends very well to a handful of models such as deepseek, claude 3.5 and gemini 1.5 flash. We may conclude that the stylometric features that the model captures are not universal across AI models and that there remains a substantial heterogeneity in writing styles. Lastly, as we may expect, the "combined" model using both TF-IDF and stylometric features perform in the middle with detection scores which are higher than or equal to the stylometric-alone model for all sources.

5.1.3 Other results from the logistic classifiers

A desirable characteristic of the logistic regression is its open-box design. We can look at what specific features of the model pushed it toward an AI or human prediction by looking at the regression coefficients associated to those features. In this set-up, when a coefficient is positive, the larger it is, the more the associated feature contributed to the model making an AI prediction. On the opposite, when a coefficient is negative, the lower it is, the more the associated features contributed to the modeling picking a human prediction. In table 6, we give the top three stylometric features that characterize human and AI writing.

The three stylistic features that characterize AI are then, in the following order, the standard deviation in word length, the number of "—" used, and the share of stopwords⁴. For humans, they are the share of proper nouns, the number of "..." used, and the lexical diversity of the text.

Table 6: Top stylometric features by predicted class

Source	Top features		
	1	2	3
AI	std_word_length	n_em_dash	stopword_share
Human	pos_share_propn	n_ellipsis	lexical_diversity

Figure 2 shows the strongest TF-IDF features associated with each class. Size of words is given by the TF-IDF weights. For the AI word cloud, we see some words that we think are typical of it. For instance, LLMs have a sycophantic behavior and tend to exaggerate some talking points even when it is justified. That is the reason for which, in our opinion, the most prominent AI-word is "important".

⁴It's actually the 4th, since the 3rd was whether the text has an —. Since it is similar to the 2nd, we preferred to show a wider diversity of informative features.



Figure 2: Word clouds based on the logistic regression trained with TF-IDF features only.

We also see a few logical connectors such as "however" and "additionally" which are indicative of more structured writing when it comes to LLMs - in line with documented work in the literature.

As for humans, it seems that our TF-IDF model has captured more pronouns such as "they" and more auxiliary verbs such as "have" and "will". This is in-line with our part-of-speech tagging results in Figure 1. Of course, our TF-IDF has also captured some artifacts of the training texts such as "venus" or "planet" which may be explained by recurring topics for the essays provided in the dataset.

5.2 DistilBERT results

Lastly, we present the results of our DistilBERT model in Table 7.

The transformer performs almost perfectly on the hold-out sample, with a recall of 99.62% and an F1-score of 99.52%, indicating that almost all AI texts are correctly discriminated. The precision rate at 99.33% shows that very few human texts are incorrectly classified. So far as the training distribution is concerned, DistilBERT has performed very well.

On the external Medium dataset, DistilBERT still has a very high AI recall rate of 99.42%. However, that seems to come at the expense of precision which decreases to 83.89%, compared with 99.33% in the hold-out sample. We conclude that either our training data was not representative enough of AI-generated texts, or that there exists too much heterogeneity in AI writing patterns to discriminate texts reliably based on this lighter transformer model alone.

Table 7: Performance of DistilBERT across evaluation datasets

Model	Hold-out sample			Medium dataset		
	F1	AI recall	Precision	F1	AI recall	Precision
DistilBERT	99.52%	99.62%	99.33%	91.97%	99.42%	83.89%

Comparison with the logistic classifiers

DistilBERT clearly performs best on the hold-out sample, with the highest F1-score, AI recall, and precision. It captures patterns in the training distribution better than the logistic classifiers. On the Medium dataset, however, its advantage disappears. Although DistilBERT keeps very high AI recall, its precision drops to 83.89%, below all logistic classifiers. This means it flags more human texts as AI, and this comes despite a very high probability threshold for prediction (around 0.85). It could be that the human essays on the external dataset resemble a bit more the AI essays on the Kaggle data. This is ultimately telling about the difficulty to detect LLMs that attempt to mimic human writing.

The combined logistic model is therefore surprisingly more balanced and slightly better overall on the external dataset.

6 Conclusion

In this report, we took on the difficult task of discriminating AI-generated texts from human writing. An initial textual analysis highlighted several markers of human and AI speeches on a stylometric level but also on the distribution of the vocabulary employed. With those insights in mind, we developed a set of logistic classifiers relying on TF-IDF features of texts, their stylometric features, or both. We found that these simple benchmarks performed very well on the hold-out sample of the training data (the "test set"), and to a lesser extent on a true out-of-sample dataset found on the internet. The open-box design of the logistic regression allowed us to discuss what were the top features selected by the models which help to discriminate between human and AI-generated texts. Lastly, we concluded our investigation by employing a pre-trained version of BERT, DistilBERT, to see how lighter versions of transformers - arguably closer to the state-of-the-art - perform on this detection task compared to the simple logistic regression. We found, surprisingly, that in the true out-of-sample dataset, it performed decently but worse than our simpler benchmarks.

7 Bibliography

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